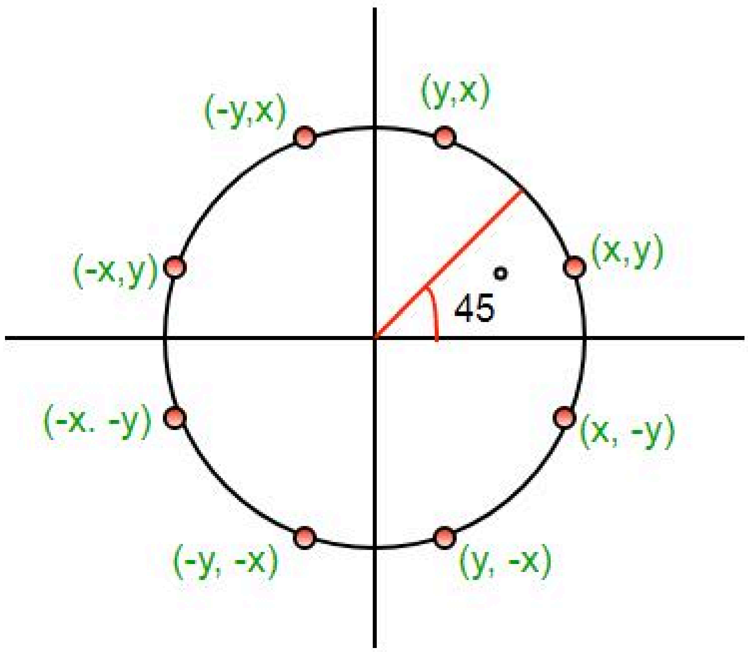
Midpoint Circle Drawing Algorithm

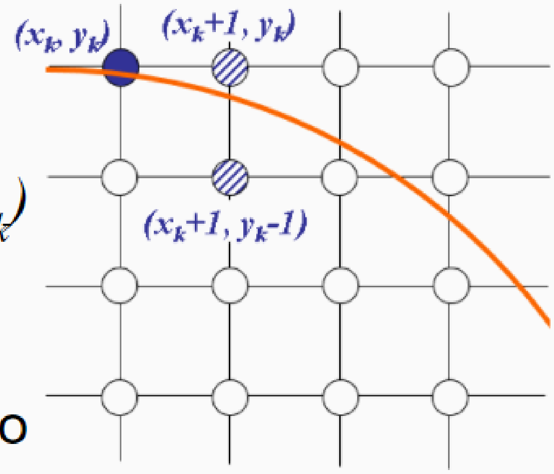
The **mid-point** circle drawing algorithm is an algorithm used to determine the points needed for rasterizing a circle. We use the **mid-point** algorithm to calculate all the perimeter points of the circle in the **first octant** and then print them along with their mirror points in the other octants. This will work because a circle is symmetric about it's centre.



Assume that we have just plotted point (x_k, y_k)

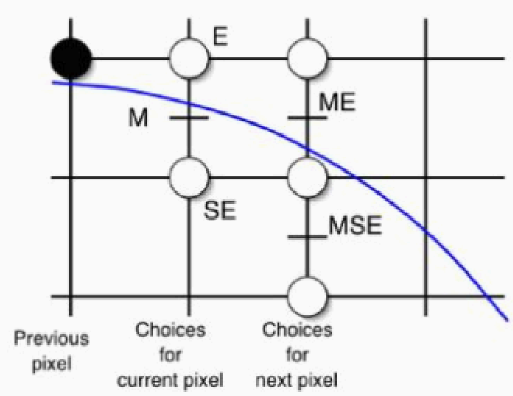
The next point is a choice between (x_k+1, y_k) and (x_k+1, y_k-1)

We would like to choose the point that is nearest to the actual circle



So how do we make this choice?

- Decision based on the midpoint of the two candidate pixels, namely E and SE //assuming 2nd octant
- Find on what side of the circle the midpoint M is
- If in, E is closer to the circle (choose E)
- If out, SE is closer to the circle (choose SE)



$$f(x, y) = x^2 + y^2 - r^2 \quad \begin{bmatrix} < 0 \text{ for } (x, y) \text{inside the circle} \\ = 0 \text{ for } (x, y) \text{on the circle} \\ > 0 \text{ for } (x, y) \text{outside the circle} \end{bmatrix} \text{.....equation 1}$$

Procedure-

Given-

Centre point of Circle = (X_0, Y_0)
Radius of Circle = R

The points generation using Mid Point Circle Drawing Algorithm involves the following steps-

Step-01:

Assign the starting point coordinates (X_0, Y_0) as-

- $X_0 = 0$
- $Y_0 = R$

Step-02:

Calculate the value of initial decision parameter P_0 as-

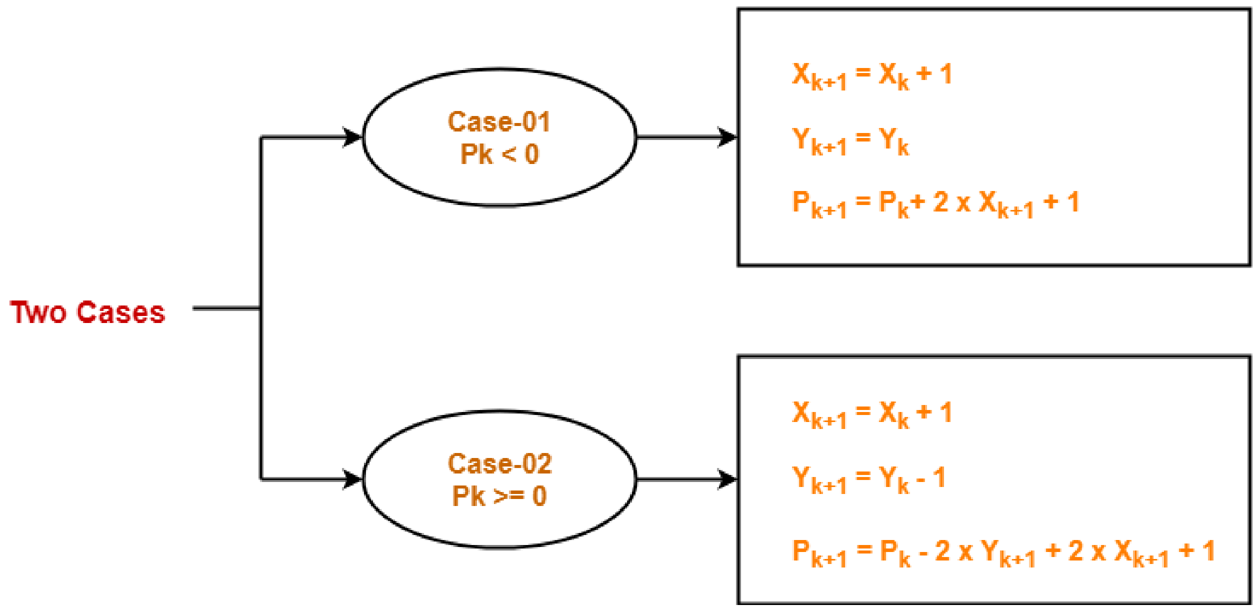
$$P_0 = 1 – R$$

Step-03:

Suppose the current point is (X_k, Y_k) and the next point is (X_{k+1}, Y_{k+1}).

Find the next point of the first octant depending on the value of decision parameter P_k.

Follow the below two cases-



Step-04:

If the given centre point (X_0, Y_0) is not (0, 0), then do the following and plot the point-

- $X_{plot} = X_c + X_0$
- $Y_{plot} = Y_c + Y_0$

Here, (X_c, Y_c) denotes the current value of X and Y coordinates.

Step-05:

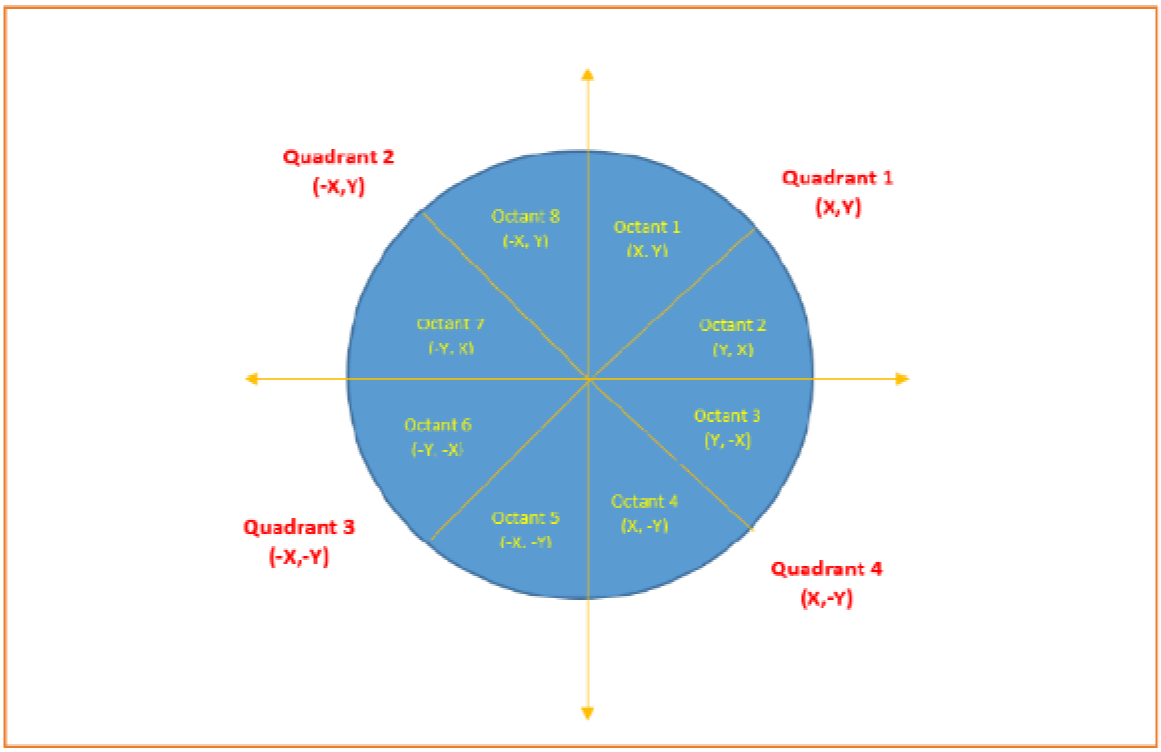
Keep repeating Step-03 and Step-04 until X_{plot} >= Y_{plot}.

Step-06:

Step-05 generates all the points for one octant.

To find the points for other seven octants, follow the eight symmetry property of circle.

This is depicted by the following figure-



Problem-01:

Given the centre point coordinates (0, 0) and radius as 10, generate all the points to form a circle.

Solution-

Given-

- Centre Coordinates of Circle $(X_0, Y_0) = (0, 0)$
- Radius of Circle = 10

Step-01:

Assign the starting point coordinates (X_0, Y_0) as-

- $X_0 = 0$
- $Y_0 = R = 10$

Step-02:

Calculate the value of initial decision parameter P_0 as-

$P_0 = 1 – R$

$P_0 = 1 – 10$

$P_0 = -9$

Step-03:

As $P_{initial} < 0$, so case-01 is satisfied.

Thus,

- $X_{k+1} = X_k + 1 = 0 + 1 = 1$
- $Y_{k+1} = Y_k = 10$
- $P_{k+1} = P_k + 2 * X_{k+1} + 1 = -9 + (2 \times 1) + 1 = -6$

Step-04:

This step is not applicable here as the given centre point coordinates is (0, 0).

Step-05:

Step-03 is executed similarly until $X_{k+1} >= Y_{k+1}$ as follows-

P_k	P_{k+1}	(X_{k+1}, Y_{k+1})
		(0, 10)
-9	-6	(1, 10)
-6	-1	(2, 10)
-1	6	(3, 10)
6	-3	(4, 9)
-3	8	(5, 9)
8	5	(6, 8)
Algorithm Terminates **These are all points for Octant-1.**		

Algorithm calculates all the points of octant-1 and terminates.

Now, the points of octant-2 are obtained using the mirror effect by swapping X and Y coordinates.

Octant-1 Points	Octant-2 Points
(0, 10)	(8, 6)
(1, 10)	(9, 5)
(2, 10)	(9, 4)
(3, 10)	(10, 3)
(4, 9)	(10, 2)
(5, 9)	(10, 1)
(6, 8)	(10, 0)
These are all points for Quadrant-1.	

Now, the points for rest of the part are generated by following the signs of other quadrants.

The other points can also be generated by calculating each octant separately.

Here, all the points have been generated with respect to quadrant-1-

Quadrant-1 (X,Y)	Quadrant-2 (-X,Y)	Quadrant-3 (-X,-Y)	Quadrant-4 (X,-Y)
(0, 10)	(0, 10)	(0, -10)	(0, -10)
(1, 10)	(-1, 10)	(-1, -10)	(1, -10)
(2, 10)	(-2, 10)	(-2, -10)	(2, -10)
(3, 10)	(-3, 10)	(-3, -10)	(3, -10)
(4, 9)	(-4, 9)	(-4, -9)	(4, -9)
(5, 9)	(-5, 9)	(-5, -9)	(5, -9)
(6, 8)	(-6, 8)	(-6, -8)	(6, -8)
(8, 6)	(-8, 6)	(-8, -6)	(8, -6)
(9, 5)	(-9, 5)	(-9, -5)	(9, -5)
(9, 4)	(-9, 4)	(-9, -4)	(9, -4)
(10, 3)	(-10, 3)	(-10, -3)	(10, -3)
(10, 2)	(-10, 2)	(-10, -2)	(10, -2)
(10, 1)	(-10, 1)	(-10, -1)	(10, -1)
(10, 0)	(-10, 0)	(-10, 0)	(10, 0)
These are all points of the Circle.			

Problem-02:

Given the centre point coordinates (4, -4) and radius as 10, generate all the points to form a circle.

Solution-

Given-

- Centre Coordinates of Circle $(X_0, Y_0) = (4, -4)$
- Radius of Circle = 10

As stated in the algorithm,

- We first calculate the points assuming the centre coordinates is (0, 0).
- At the end, we translate the circle.

Step-01, Step-02 and Step-03 are already completed in Problem-01.

Now, we find the values of X_{plot} and Y_{plot} using the formula given in Step-04 of the main algorithm.

The following table shows the generation of points for Quadrant-1-

- $X_{plot} = X_c + X_0 = 4 + X_0$
- $Y_{plot} = Y_c + Y_0 = -4 + Y_0$

(X_{k+1}, Y_{k+1})	(X_{plot}, Y_{plot})
(0, 10)	(4, 14)
(1, 10)	(5, 14)
(2, 10)	(6, 14)
(3, 10)	(7, 14)
(4, 9)	(8, 13)
(5, 9)	(9, 13)
(6, 8)	(10, 12)
(8, 6)	(12, 10)
(9, 5)	(13, 9)
(9, 4)	(13, 8)
(10, 3)	(14, 7)
(10, 2)	(14, 6)
(10, 1)	(14, 5)
(10, 0)	(14, 4)
These are all points for Quadrant-1.	

The following table shows the points for all the quadrants-

Q1	Q1-Q1(C)	Q2	Q2-Q2(C)	Q3	Q3-Q3(C)	Q4	Q4-Q4(C)
	(Xc,Yc)=(4,-4) X0+Xc=x1, X0+Yc=Y1		(Xc,Yc)=(4,-4) X0+Xc=x1, X0+Yc=Y1		(Xc,Yc)=(4,-4) X0+Xc=x1, X0+Yc=Y1		(Xc,Yc)=(4,-4) X0+Xc=x1, X0+Yc=Y1
(0,10)	(4,6)	(0,10)	(4,6)	(0,-10)	(4,-14)	(0,-10)	(4,-14)
(1,10)	(5,6)	(-1,10)	(3,6)	(-1,-10)	(3,-14)	(1,-10)	(5,-14)
(2,10)	(6,6)	(-2,10)	(2,6)	(-2,-10)	(2,-14)	(2,-10)	(6,-14)
(3,10)	(7,6)	(-3,10)	(1,6)	(-3,-10)	(1,-14)	(3,-10)	(7,-14)
(4,9)	(8,5)	(-4,9)	(0,5)	(-4,-9)	(0,-13)	(4,-9)	(8,-13)
(5,9)	(9,5)	(-5,9)	(-1,5)	(-5,-9)	(-1,-13)	(5,-9)	(9,-13)
(6,8)	(10,4)	(-6,8)	(-2,4)	(-6,-8)	(-2,-12)	(6,-8)	(10,-12)
(7,7)	(11,3)	(-7,7)	(-3,3)	(-7,-7)	(-3,-11)	(7,-7)	(11,-11)
(8,6)	(12,2)	(-8,6)	(-4,2)	(-8,-6)	(-4,-10)	(8,-6)	(12,-10)
(9,5)	(13,1)	(-9,5)	(-5,1)	(-9,-5)	(-5,-9)	(9,-5)	(13,-9)
(9,4)	(13,0)	(-9,4)	(-5,0)	(-9,-4)	(-5,-8)	(9,-4)	(14,-8)
(10,3)	(14,-1)	(-10,3)	(-6,-1)	(-10,-3)	(-6,-7)	(10,-3)	(14,-7)
(10,2)	(14,-2)	(-10,2)	(-6,-2)	(-10,-2)	(-6,-6)	(10,-2)	(14,-6)
(10,1)	(14,-3)	(-10,1)	(-6,-3)	(-10,-1)	(-6,-5)	(10,-1)	(14,-5)
(10,0)	(14,-4)	(-10,0)	(-6,-4)	(-10,0)	(-6,-4)	(10,0)	(14,-4)

Advantages of Mid Point Circle Drawing Algorithm-

The advantages of Mid Point Circle Drawing Algorithm are-

- It is a powerful and efficient algorithm.
- The entire algorithm is based on the simple equation of circle $X^2 + Y^2 = R^2$.
- It is easy to implement from the programmer's perspective.
- This algorithm is used to generate curves on raster displays.

Disadvantages of Mid Point Circle Drawing Algorithm-

The disadvantages of Mid Point Circle Drawing Algorithm are-

- Accuracy of the generating points is an issue in this algorithm.
- The circle generated by this algorithm is not smooth.
- This algorithm is time consuming.